CHAPTER 11 | Customer Relationship Management and Supply Chain Management

CHAPTER ROADMAP

CRM and SCM coordinate relationships and flows beyond isolated departments.

Learning outcomes

1. Explain and apply: CRM foundations.
2. Explain and apply: Customer lifecycle.
3. Explain and apply: Supply chain structure.
4. Explain and apply: Inventory and bullwhip.
5. Explain and apply: Digital integration.
6. Explain and apply: Resilience and sustainability.

Concept map

BUSINESS QUESTION → DATA / PEOPLE / PROCESS → INFORMATION SYSTEM → ACTION → MEASURABLE OUTCOME

How to use this chapter

Read the six topic pages, explain the definitions aloud, study the scenario, and answer the questions without looking. Then check the model answers. This guide is original educational commentary, not a reproduction of the textbook.

Big picture

CRM and SCM coordinate relationships and flows beyond isolated departments. The central skill is to connect terminology to a real organisational decision. Whenever you meet a new concept, identify its purpose, required inputs, expected benefits, limitations, and the people responsible for it.

CRM foundations

Core explanation

CRM combines customer strategy, processes and data across marketing, sales and service. A single customer view helps but depends on lawful, accurate data.

Deeper understanding

To apply crm foundations, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• CRM: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• customer touchpoint: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• customer 360: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

Support agents see recent orders before responding.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of crm foundations in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Customer lifecycle

Core explanation

Acquisition, onboarding, retention and recovery require different activities. Customer lifetime value estimates economic contribution but should not erase fairness considerations.

Deeper understanding

To apply customer lifecycle, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• retention: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• churn: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• CLV: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A subscription service contacts customers showing signs of cancellation.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of customer lifecycle in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Supply chain structure

Core explanation

Suppliers, manufacturers, distributors, retailers and customers exchange goods, money and information. Coordination reduces delays and uncertainty.

Deeper understanding

To apply supply chain structure, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• supply chain: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• SCM: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• logistics: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A manufacturer shares forecasts with key suppliers.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of supply chain structure in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Inventory and bullwhip

Core explanation

Small changes in consumer demand can create larger upstream order fluctuations. Better visibility and stable replenishment policies reduce amplification.

Deeper understanding

To apply inventory and bullwhip, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• bullwhip effect: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• safety stock: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• lead time: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

Retailers over-order after a temporary sales spike.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of inventory and bullwhip in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Digital integration

Core explanation

ERP, CRM, warehouse systems and supplier portals exchange data. Standards, APIs and data quality are essential for visibility.

Deeper understanding

To apply digital integration, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• integration: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• EDI: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• supplier portal: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A shipment scan updates estimated delivery for customers.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of digital integration in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Resilience and sustainability

Core explanation

Diversification, contingency plans and supplier evaluation improve resilience. Balance cost efficiency with ethical sourcing and environmental impacts.

Deeper understanding

To apply resilience and sustainability, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• resilience: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• traceability: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• supplier risk: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A firm qualifies alternative suppliers for critical parts.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of resilience and sustainability in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

CHAPTER 12 | Business Analytics

CHAPTER ROADMAP

Analytics combines data, questions and methods to support evidence-based decisions.

Learning outcomes

1. Explain and apply: Analytics questions.

2. Explain and apply: Data preparation.

3. Explain and apply: Exploration and visualisation.

4. Explain and apply: Predictive modelling.

5. Explain and apply: Prescriptive decisions.

6. Explain and apply: Responsible analytics.

Concept map

BUSINESS QUESTION → DATA / PEOPLE / PROCESS → INFORMATION SYSTEM → ACTION → MEASURABLE OUTCOME

How to use this chapter

Read the six topic pages, explain the definitions aloud, study the scenario, and answer the questions without looking. Then check the model answers. This guide is original educational commentary, not a reproduction of the textbook.

Big picture

Analytics combines data, questions and methods to support evidence-based decisions. The central skill is to connect terminology to a real organisational decision. Whenever you meet a new concept, identify its purpose, required inputs, expected benefits, limitations, and the people responsible for it.

Analytics questions

Core explanation

Start with a business problem and measurable outcome. Distinguish descriptive, diagnostic, predictive and prescriptive questions before choosing tools.

Deeper understanding

To apply analytics questions, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• descriptive: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• diagnostic: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• predictive: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• prescriptive: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A retailer asks why returns rose and how to reduce them.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of analytics questions in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Data preparation

Core explanation

Collect, join, validate and clean data; document missing values, definitions and biases. Poor inputs produce misleading models and dashboards.

Deeper understanding

To apply data preparation, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• data cleaning: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• missing data: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• bias: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

Different departments use inconsistent definitions of active customer.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of data preparation in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Exploration and visualisation

Core explanation

Charts and summary statistics reveal patterns, distributions and anomalies. Use appropriate scales and avoid confusing correlation with causation.

Deeper understanding

To apply exploration and visualisation, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• dashboard: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• KPI: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• correlation: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A line chart shows monthly revenue trends and seasonality.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of exploration and visualisation in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Predictive modelling

Core explanation

Classification estimates categories; regression estimates values; forecasts predict future time-based outcomes. Test performance on unseen data.

Deeper understanding

To apply predictive modelling, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• classification: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• regression: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• forecast: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• overfitting: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A model estimates which invoices may be paid late.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of predictive modelling in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Prescriptive decisions

Core explanation

Optimisation and simulation compare actions under constraints. Recommendations require assumptions, costs and human judgement.

Deeper understanding

To apply prescriptive decisions, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• optimisation: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• simulation: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• constraint: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A delivery planner allocates vehicles to minimise distance.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of prescriptive decisions in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Responsible analytics

Core explanation

Monitor model drift, privacy, fairness and explainability. Present uncertainty and limitations instead of false precision.

Deeper understanding

To apply responsible analytics, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• model drift: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• fairness: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• explainability: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A lending model is checked for uneven errors across groups.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of responsible analytics in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

CHAPTER 13 | Acquiring Information Systems and Applications

CHAPTER ROADMAP

Successful acquisition aligns business needs, technical choices and organisational change.

Learning outcomes

1. Explain and apply: Requirements and scope.
2. Explain and apply: Feasibility analysis.
3. Explain and apply: Build, buy or outsource.
4. Explain and apply: Development life cycle.
5. Explain and apply: Agile and quality assurance.
6. Explain and apply: Deployment and change.

Concept map

BUSINESS QUESTION → DATA / PEOPLE / PROCESS → INFORMATION SYSTEM → ACTION → MEASURABLE OUTCOME

How to use this chapter

Read the six topic pages, explain the definitions aloud, study the scenario, and answer the questions without looking. Then check the model answers. This guide is original educational commentary, not a reproduction of the textbook.

Big picture

Successful acquisition aligns business needs, technical choices and organisational change. The central skill is to connect terminology to a real organisational decision. Whenever you meet a new concept, identify its purpose, required inputs, expected benefits, limitations, and the people responsible for it.

Requirements and scope

Core explanation

Define stakeholders, functional and non-functional requirements, constraints and success criteria. Prioritise needs before selecting a solution.

Deeper understanding

To apply requirements and scope, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• requirements: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• scope: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• acceptance criteria: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A library needs searchable inventory and reliable accessibility.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of requirements and scope in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Feasibility analysis

Core explanation

Assess technical, economic, organisational, legal and schedule feasibility. Include lifetime costs, training and risk, not only purchase price.

Deeper understanding

To apply feasibility analysis, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• feasibility: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• TCO: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• cost-benefit: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A small company compares hosted and self-managed software.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of feasibility analysis in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Build, buy or outsource

Core explanation

Custom development offers fit and control but increases maintenance demands. Packages accelerate deployment but constrain processes; outsourcing introduces dependency.

Deeper understanding

To apply build, buy or outsource, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• build vs buy: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• outsourcing: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• SaaS: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A startup chooses a hosted CRM rather than developing one.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of build, buy or outsource in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Development life cycle

Core explanation

Planning, analysis, design, implementation, testing and maintenance form an iterative or staged life cycle. Feedback may require revisiting earlier decisions.

Deeper understanding

To apply development life cycle, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• SDLC: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• waterfall: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• iterative development: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A team prototypes an appointment system with real users.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of development life cycle in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Agile and quality assurance

Core explanation

Agile delivers small increments and incorporates feedback. Testing includes unit, integration, system, acceptance, security and usability checks.

Deeper understanding

To apply agile and quality assurance, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• agile: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• sprint: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• UAT: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• regression testing: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

Students test a registration flow before release.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of agile and quality assurance in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Deployment and change

Core explanation

Migration, training, rollout strategy, monitoring and support determine adoption. A technically correct product may still fail operationally.

Deeper understanding

To apply deployment and change, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• migration: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• cutover: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• change management: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A hospital pilots a new system in one department first.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of deployment and change in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

CHAPTER 14 | Artificial Intelligence

CHAPTER ROADMAP

AI supports perception, prediction and decision-making but requires careful evaluation and human accountability.

Learning outcomes

1. Explain and apply: AI landscape.
2. Explain and apply: Learning approaches.
3. Explain and apply: Language and perception.
4. Explain and apply: Automation and applications.
5. Explain and apply: Evaluation and limitations.
6. Explain and apply: Responsible governance.

Concept map

BUSINESS QUESTION → DATA / PEOPLE / PROCESS → INFORMATION SYSTEM → ACTION → MEASURABLE OUTCOME

How to use this chapter

Read the six topic pages, explain the definitions aloud, study the scenario, and answer the questions without looking. Then check the model answers. This guide is original educational commentary, not a reproduction of the textbook.

Big picture

AI supports perception, prediction and decision-making but requires careful evaluation and human accountability. The central skill is to connect terminology to a real organisational decision. Whenever you meet a new concept, identify its purpose, required inputs, expected benefits, limitations, and the people responsible for it.

AI landscape

Core explanation

AI is the broad field; machine learning learns patterns from data; deep learning uses layered neural networks. Rule-based systems use explicit logic instead.

Deeper understanding

To apply ai landscape, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• AI: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• ML: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• deep learning: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• expert system: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A fraud model learns patterns while a rule blocks expired cards.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of ai landscape in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Learning approaches

Core explanation

Supervised learning uses labelled examples; unsupervised learning explores structure; reinforcement learning uses rewards to guide behaviour. Match method to problem.

Deeper understanding

To apply learning approaches, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• supervised: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• unsupervised: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• reinforcement learning: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A classifier learns from labelled spam emails.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of learning approaches in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Language and perception

Core explanation

NLP processes text and speech; computer vision analyses images. Generative AI creates content but may produce plausible inaccuracies.

Deeper understanding

To apply language and perception, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• NLP: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• computer vision: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• generative AI: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• hallucination: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A chatbot drafts support replies that staff review.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of language and perception in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Automation and applications

Core explanation

AI supports recommendations, forecasting, anomaly detection and robotics. RPA automates repetitive rules and does not inherently require learning.

Deeper understanding

To apply automation and applications, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• recommendation system: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• anomaly detection: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• RPA: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A service desk routes tickets based on message content.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of automation and applications in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Evaluation and limitations

Core explanation

Measure accuracy, precision, recall, robustness and operating costs as appropriate. Data shifts, weak labels and overfitting can undermine performance.

Deeper understanding

To apply evaluation and limitations, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• precision: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• recall: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• robustness: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• overfitting: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A medical screening tool prioritises sensitivity and careful validation.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of evaluation and limitations in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Responsible governance

Core explanation

Protect privacy, examine bias, document intended use and keep human oversight for consequential decisions. Test security and monitor real-world impacts.

Deeper understanding

To apply responsible governance, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• fairness: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• transparency: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• human oversight: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A recruitment tool is audited and allows human appeal.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of responsible governance in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Sources and scope

Rainer, R. Kelly and Brad Prince (2022), Introduction to Information Systems, 9th ed., International Adaptation. Chapter titles verified against the Wiley publisher table of contents. Original study notes; supplement with the UOC assigned readings and official chapter case studies.